

AI & Search Algorithms Project Report

Lab Solutions & Algorithm Evaluation | PyCharm Project Documentation

Problem #	Problem Name	Algorithm Applied	Optimal Path / Output	Total Cost / Result
Problem 1	Doctor Shift Scheduling	Backtracking Search (CSP)	D1: Morning, D2: Afternoon, D3: Night	Valid Schedule
Problem 2	5x5 Grid Pathfinding	A* Search (f = g + h)	(0,0) → (1,0) → (2,0) → (3,0) → (3,1) → (3,2) → (4,2) → (4,3) → (4,4)	8 Steps
Problem 3	Autonomous Rescue Robot	Online Search (LRTA*)	Dynamic Replanning Path (Senses adjacent cells)	Optimal Cost Path

1. Doctor Shift Scheduling (Backtracking CSP)

Objective: Assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) without conflict.
Constraints: (i) D1 ≠ Night, (ii) D2 < D3, (iii) AllDiff, (iv) D3 ≠ Morning, (v) Exactly 1 shift per doctor.
Results: Backtracking search prunes invalid branches early and identifies the sole valid schedule: **D1 = Morning, D2 = Afternoon, D3 = Night.**

2. 5x5 Grid Pathfinding (A* Search)

Objective: Find optimal path from (0,0) to (4,4) avoiding obstacles at (1,1), (1,2), (2,1), (3,3).
Evaluation Function: $f(n) = g(n) + h(n)$, where $g(n)$ is step cost (1) and $h(n)$ is Manhattan distance.
Results: A* Search evaluates nodes in priority order, exploring 9 states to reach the goal with an optimal cost of **8 steps.**

3. Autonomous Rescue Robot (Online Search / LRTA*)

PEAS Category	Description
Performance	Minimize path cost (Clear=1, Risky=3) while locating survivor safely.
Environment	Partially observable 5x5 grid, dynamic risk zones, static obstacles.
Actuators	4-directional movement motors (Up, Down, Left, Right).
Sensors	Short-range proximity & hazard detection sensors for adjacent cells.